MULTI LINEAR REGRESSION

Multi Linear Regression

Here there is

- One dependent variable
- Multiple independent variables

For Example:

Features

Area	Bedrooms	Age	Price
2600	3	20	550000
3000	4	15	565000
3200	4	18	610000
3600	3	30	595000
4000	5	8	760000

Here:

- Price is the dependent variable
- Area, Bedroom and Age are the independent variables
- Area, Bedroom and Age are also called attributes or features

The equation for Multi Linear Regression is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \cdots \ldots\ldots\ldots + \beta_n X_n + \epsilon$$

Where

Y is the dependent variable

$X_1, X_2, X_3, \ldots\ldots\ldots\ldots X_n$ are the independent variables

β_0 is the Y intercept

$\beta_1, \beta_2, \beta_3, \ldots\ldots\ldots\ldots \beta_n$ are the coefficients of X or the slopes

ϵ is the error factor

Finding the Coefficients:

The equation is:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

So, Given the Data:

	Y	X1	X2	X3
0	43	30	63	33
1	63	45	47	52
2	71	68	67	62
3	61	46	83	42
4	81	66	84	42

We need to find the best fit line using the dependent and independent variables.

So, We need to solve:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

$$We\ need\ to\ find\ \beta_0, \beta_1, \beta_2, \beta_3$$

Y is known

Y
43
63
71
61
81
43

X is known

X0	X1	X2	X3
1	30	63	33
1	45	47	52
1	68	67	62
1	46	83	42
1	66	84	42
1	36	50	66

Now let us solve the equation:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$X^T = \begin{matrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 30 & 45 & 68 & 46 & 66 & 36 \\ 63 & 47 & 67 & 83 & 84 & 50 \\ 33 & 52 & 62 & 42 & 42 & 66 \end{matrix}$$

$X^T X =$

$$X^T = \begin{matrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 30 & 45 & 68 & 46 & 66 & 36 \\ 63 & 47 & 67 & 83 & 84 & 50 \\ 33 & 52 & 62 & 42 & 42 & 66 \end{matrix} \quad X = \begin{matrix} 1 & 30 & 63 & 33 \\ 1 & 45 & 47 & 52 \\ 1 & 68 & 67 & 62 \\ 1 & 46 & 83 & 42 \\ 1 & 66 & 84 & 42 \\ 1 & 36 & 50 & 66 \end{matrix}$$

$$X^T X = \begin{matrix} 6 & 291 & 394 & 297 \\ 291 & 15317 & 19723 & 14626 \\ 394 & 19723 & 27112 & 18991 \\ 297 & 14626 & 18991 & 15521 \end{matrix}$$

$$X^T X =$$

6	291	394	297
291	15317	19723	14626
394	19723	27112	18991
297	14626	18991	15521

$$(X^T X)^{-1} =$$

16.95802	0.083191	-0.17313	-0.19106
0.083191	0.001894	-0.00155	-0.00148
-0.17313	-0.00155	0.002356	0.001891
-0.19106	-0.00148	0.001891	0.002802

$$X^T = \begin{matrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 30 & 45 & 68 & 46 & 66 & 36 \\ 63 & 47 & 67 & 83 & 84 & 50 \\ 33 & 52 & 62 & 42 & 42 & 66 \end{matrix}$$

$$Y = \begin{matrix} 43 \\ 63 \\ 71 \\ 61 \\ 81 \\ 43 \end{matrix}$$

$$X^T Y = \begin{matrix} 362 \\ 18653 \\ 24444 \\ 17899 \end{matrix}$$

Now Solving the Equation

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$(X^T X)^{-1} = \begin{matrix} 16.95802 & 0.083191 & -0.17313 & -0.19106 \\ 0.083191 & 0.001894 & -0.00155 & -0.00148 \\ -0.17313 & -0.00155 & 0.002356 & 0.001891 \\ -0.19106 & -0.00148 & 0.001891 & 0.002802 \end{matrix}$$

$$X^T Y = \begin{matrix} 362 \\ 18653 \\ 24444 \\ 17899 \end{matrix}$$

Therefore:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$\hat{\beta} = \begin{matrix} 38.8731 \\ 1.062949 \\ -0.15181 \\ -0.40655 \end{matrix}$$

That is: The intercept and the Coefficients are:

$$\beta_0 = 38.8731 \quad \beta_1 = 1.062949 \quad \beta_2 = -0.15181 \quad \beta_3 = -0.40655$$

Therefore for a new set of Features:

Y	X1	X2	X3
?	39	59	40

Applying:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

$$Y = 38.8731 + (1.062949 * 39) + (-0.15181 * 59) + (-0.40655 * 40)$$

$$Y = 55.10958$$

Mean Squared Error:

	Y	X1	X2	X3
0	43	30	63	33
1	63	45	47	52
2	71	68	67	62
3	61	46	83	42
4	81	66	84	42

Mean Squared Error:

$$Y_{p1} = 38.8731 + (1.062949 * 43) + (-0.15181 * 30) + (-0.40655 * 63)$$

$$Y_{p2} = 38.8731 + (1.062949 * 63) + (-0.15181 * 45) + (-0.40655 * 47)$$

$$Y_{p3} = 38.8731 + (1.062949 * 71) + (-0.15181 * 68) + (-0.40655 * 67)$$

$$Y_{p4} = 38.8731 + (1.062949 * 61) + (-0.15181 * 46) + (-0.40655 * 83)$$

$$Y_{p5} = 38.8731 + (1.062949 * 81) + (-0.15181 * 66) + (-0.40655 * 84)$$

Mean Squared Error:

Y	Yp	Error(Y-Yp)	Error Squared
43	47.78166	-4.78166	22.86429328
63	58.43034	4.569658	20.88177075
71	75.77656	-4.77656	22.81554469
61	58.09379	2.906214	8.446078129
81	79.20097	1.799032	3.236515623
43	42.71668	0.283321	0.080270799

Sum of Errors	78.32447328
Mean Squared Error	13.05407888

R Squared: The coefficient of Determination

$$R^2 = \frac{\sum(yp - \bar{y})^2}{\sum(y - \bar{y})^2}$$

- R-Squared value is a statistical measure of how close the data are to the fitted regression line.
- It is also known as the coefficient of determination or coefficient of multiple determination for multiple regression.
- 1(100%) is the best score and 0(0%) is worst score..
- In general it is considered that a high R-Squared value model is a good model. But you can also have a lower R-Squared value for a good model as well.

Y	Y-Y_Mean	(Y-Y_Mean)^2	Yp	Yp-Y_Mean	(Yp-Y_Mean)^2
43	-17.3333	300.4444444	47.78166219	-12.5517	157.5444
63	2.666667	7.111111111	58.43034238	-1.90299	3.621375
71	10.66667	113.7777778	75.77656202	15.44323	238.4933
61	0.666667	0.444444444	58.09378629	-2.23955	5.015571
81	20.66667	427.1111111	79.20096814	18.86763	355.9876
43	-17.3333	300.4444444	42.71667898	-17.6167	310.3465
Mean		**Sum**			**Sum**
60.33333		1149.333333			1071.009

R_Squared = 1079.009 / 1149.333333

R_Squared= 0.931852256